

Swarm Logistics & Warehouse Automation SRS Document

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1. Introduction

This project defines a decentralized multi-agent system (MAS) for warehouse automation, in which multiple autonomous robotic agents coordinate to fulfill tasks within a shared environment. The system operates without centralized control, requiring agents to make independent decisions based on local information while collectively achieving global system objectives.

The primary objective of the system is to maximize order fulfillment efficiency while ensuring safe, conflict-free operation. Agents must operate under dynamic conditions, including continuously arriving tasks and varying levels of congestion within the environment.

The system is modeled as a discrete, grid-based warehouse simulation with synchronous time steps. Agents are responsible for task allocation, navigation, and coordination while operating under partial observability, with no prior knowledge of future task arrivals.

The scope of this project is limited to the design and simulation of the multi-agent system. It includes the formal specification of environment characteristics, agent behaviors, interaction mechanisms, functional requirements, and performance evaluation criteria. Physical deployment, hardware constraints, and real-world integration are outside the scope of this work.

The system assumes a shared environment with reliable communication and deterministic action outcomes, and focuses on evaluating decentralized coordination strategies under scalable and dynamic conditions.

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2. Environment & Specification:

The system operates within a warehouse environment modeled as a discrete grid where each cell represents a navigable or occupied location. The environment defines the spatial and operational constraints under which all agents function.

2.1 Environment Type (Taxonomy)

The system operates in a warehouse environment modeled as a multi-agent system with the following characteristics:

- **Multi-agent:** Multiple autonomous robots operate concurrently in the same environment.
- **Collaborative:** All agents share the common objective of maximizing overall system throughput.
- **Partially observable:** Agents have limited knowledge of the global system state and future task arrivals.
- **Dynamic:** New tasks may be introduced during system execution.
- **Discrete:** The environment is represented as a grid with discrete spatial and temporal states.
- **Deterministic actions:** Movement actions produce predictable outcomes.
- **Stochastic task arrivals:** Tasks are generated dynamically according to a stochastic process.

2.2 Environment Model

The warehouse is modeled as a two-dimensional discrete grid of size $N \times M$.

Each grid cell can be in one of the following states:

- Free space
- Occupied by an agent
- Obstacle (static, non-traversable)
- Pickup location
- Drop-off location

Agents move between adjacent cells (up, down, left, right), subject to occupancy and collision constraints.

2.3 Temporal Model

The system operates in discrete time steps. At each time step:

1. Agents perceive their environment
2. Agents compute decisions based on current information
3. Agents execute one action simultaneously

All agents act synchronously within the simulation.

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2.4 Physical and Logical Boundaries

- The warehouse layout is fixed during simulation and includes predefined obstacle regions.
- The grid dimensions $N \times M$ are bounded and finite.
- Agents cannot move outside the grid boundaries.
- Each grid cell can be occupied by at most one agent at any given time step.
- Position swapping between two agents within the same time step is not permitted (to prevent edge conflicts).

2.5 Task Model

A task is defined as a tuple:

$$T = (L_{pickup}, L_{dropoff})$$

where:

- L_{pickup} is the pickup location
- $L_{dropoff}$ is the delivery destination

Tasks are generated dynamically during runtime and added to a global task pool. Agents select and execute tasks based on decentralized coordination mechanisms.

2.6 Observability and Communication

- Each agent has full knowledge of the static environment (grid layout and obstacle locations).
- Agents are aware of their own position and assigned tasks.
- Agents may have limited or delayed knowledge of other agents' positions, depending on communication.
- Future task arrivals are not known in advance.

Agents can communicate short-term intent information (e.g., planned movement or reservation of cells) to coordinate actions and reduce conflicts. Communication is assumed to be reliable with negligible delay.

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3. Agent Specification

3.1 Agent Types

The system consists of a set of homogeneous autonomous agents, each representing a warehouse robot. All agents share identical capabilities, decision-making logic, and objectives. There is no centralized controller; coordination is achieved through decentralized interactions among agents.

3.2 Agent Goals

Each agent aims to contribute to the global system objective of maximizing throughput while ensuring safe and efficient operation. Specifically, each agent seeks to:

- Successfully complete assigned tasks (pickup and delivery)
- Minimize task completion time
- Avoid collisions with other agents
- Avoid contributing to congestion and deadlocks
- Maintain continuous utilization when tasks are available

Agent decisions are made locally but are expected to collectively improve overall system performance.

3.3 Perception Capabilities

Each agent has access to the following information:

- Its current position within the grid
- The static environment layout (grid structure and obstacle locations)
- Its currently assigned task (if any)
- The positions or intended movements of nearby agents (through communication or sensing)
- The set of available tasks in the system (if globally visible)

Agents do not have knowledge of future task arrivals.

3.4 Action Capabilities

At each time step, an agent may perform one of the following actions:

- Move to an adjacent grid cell (up, down, left, right), subject to occupancy constraints
- Wait (remain in the current cell)
- Pick up an item at a designated pickup location
- Deliver an item at a designated drop-off location
- Participate in task allocation (e.g., submit bids or accept assignments)
- Communicate short-term intent or coordination information to other agents

All actions are executed within the constraints defined by the environment (e.g., no collisions, valid movement).

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4. Functional Requirements

4.1 System-Level Requirements

FR-1: Task Generation

The system SHALL generate tasks dynamically during runtime, where each task consists of a pickup and drop-off location.

FR-2: Task Pool Management

The system SHALL maintain a pool of pending tasks. Each task SHALL exist in one of the following states: unassigned, assigned, or completed.

FR-3: Task Assignment Consistency

The system SHALL ensure that each task is assigned to at most one agent at any given time.

FR-4: Task Lifecycle Management

The system SHALL update task states as follows:

- A task transitions from unassigned to assigned when selected by an agent
- A task transitions from assigned to completed upon successful delivery
- Completed tasks SHALL be removed from the active task pool

FR-5: Collision Avoidance

The system SHALL ensure that no two agents occupy the same grid cell at the same time step.

FR-6: Movement Validity

The system SHALL restrict agent movements to valid adjacent grid cells and prevent movement into obstacles or outside the grid boundaries.

FR-7: Deadlock Detection and Resolution

The system SHALL detect deadlock situations and SHALL ensure that agents involved in a deadlock eventually resume progress toward their goals.

FR-8: Congestion Handling

The system SHALL enable agents to adjust behavior in response to congestion in shared areas to maintain system throughput.

FR-9: Task Completion Tracking

The system SHALL track and record the completion of tasks by agents.

FR-10: System Continuity

The system SHALL continue operating as long as tasks are available or generated, without requiring centralized intervention.

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4.2 Agent-Level Requirements

FR-11: Task Allocation Participation

Agents SHALL participate in a decentralized task allocation mechanism to select tasks from the available task pool.

FR-12: Single Task Execution

An agent SHALL handle at most one task at a time.

FR-13: Task Execution

An agent assigned to a task SHALL navigate from its current position to the task's pickup location and subsequently to the drop-off location.

FR-14: Path Planning

Agents SHALL compute valid paths that respect environmental constraints and avoid conflicts with other agents.

FR-15: Dynamic Replanning

Agents SHALL replan their paths when encountering blocked routes, congestion, or conflicts with other agents.

FR-16: Collision Avoidance Behavior

Agents SHALL adjust their movement decisions to avoid collisions with other agents.

FR-17: Congestion-Aware Behavior

Agents SHALL consider congestion or predicted conflicts when selecting or updating movement decisions.

FR-18: Idle Behavior

Agents without assigned tasks SHALL remain idle or position themselves in a way that does not block active agents.

FR-19: Communication Usage

Agents SHALL use communicated intent or coordination information when making movement decisions.

FR-20: Task Completion Confirmation

Agents SHALL signal task completion upon successful delivery to the drop-off location.

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5. Simulation & Visualization Requirements

5.1 Simulation Environment

The system SHALL include a simulation platform that models the warehouse environment and executes agent behaviors in real time using discrete time steps.

The simulator SHALL support multiple agents operating concurrently within the grid-based environment defined in Section 2.

The simulator SHALL support deterministic execution through the use of a configurable random seed to enable reproducibility of experiments.

5.2 Visualization Requirements

The simulator SHALL provide a real-time graphical visualization of the environment, including:

- The grid layout, including obstacles, pickup locations, and drop-off locations
- The position of each agent at every time step
- The current task assignment of each agent
- The planned or reserved paths of agents (if applicable)
- Areas of high congestion (e.g., visually highlighted regions)

Each agent SHALL be uniquely identifiable (e.g., by color or ID).

5.3 State and Data Visualization

The simulator SHALL display relevant system and agent-level state information, including:

- Current simulation time step
- Number of active agents
- Number of pending, assigned, and completed tasks
- Task assignments per agent
- Agent states (idle, moving, executing task)
- Detected conflicts or blocked states

The system SHALL log key events, including:

- Task assignment events
- Task completion events
- Collision attempts or prevented conflicts
- Deadlock detection and resolution events

The system SHALL ensure that logs contain sufficient information to reconstruct the sequence of agent decisions and system states.

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5.4 Interaction and Control

The simulator SHALL provide user interaction capabilities, including:

- Start, pause, and reset simulation
- Adjustable simulation speed
- Step-by-step execution for detailed inspection
- Ability to configure initial parameters, including number of agents and task generation rate

5.5 Verification and Observability

The simulator SHALL enable an observer to verify that system requirements are satisfied by providing:

- Visual confirmation that no two agents occupy the same cell simultaneously (FR-5)
- Visualization of agent paths and decisions to verify path planning and replanning behavior (FR-14, FR-15)
- Indicators of congestion and agent responses to it (FR-8, FR-17)
- Explicit visualization or logging of deadlock situations and their resolution (FR-7)
- Tracking of task lifecycle (unassigned → assigned → completed) (FR-2, FR-4)

The simulator SHALL provide clear visual or logged indicators of any violations of system constraints (e.g., collision attempts, invalid movements).

The simulator SHALL ensure that all functional requirements defined in Section 4 can be validated through either visualization or logged data.

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6. Interaction Models

6.1 Coordination Model

The system employs a fully decentralized coordination model in which agents independently make decisions based on local information and communication with other agents.

No centralized controller is responsible for task allocation or movement coordination.

6.2 Task Allocation Mechanism

Task allocation is performed using a decentralized auction-based mechanism:

1. When a new task becomes available, it is announced to all agents or made visible in the shared task pool.
2. Agents independently evaluate the task based on their local state (e.g., position, workload).
3. Agents submit bids representing their suitability for executing the task.
4. Each agent determines the winning bid using a consistent selection rule applied locally.
5. In the case of equal bids, a deterministic tie-breaking rule SHALL be applied (e.g., lowest agent ID).
6. The selected agent assigns the task to itself and updates the task state to “assigned.”

The system SHALL ensure that each task is eventually assigned to an agent, provided agents are available.

6.3 Communication Model

Agents communicate coordination-related information to reduce conflicts and improve efficiency.

The system SHALL support the following message types:

- **Task-related messages:** task announcements and assignment confirmations
- **Intent messages:** short-term planned movements or reservations of future positions
- **Conflict-related messages:** notifications of detected conflicts or blocked states

Communication is assumed to be reliable, with negligible delay. Messages may be broadcast globally or shared locally among nearby agents, depending on the simulation configuration.

6.4 Movement Coordination

Agents coordinate their movements through the exchange of intent information:

- Agents SHALL broadcast short-term movement intentions (e.g., next position or planned path segment)
- Agents SHALL consider received intent information when selecting actions
- Agents SHALL adjust movement decisions to avoid conflicts based on shared information

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6.5 Conflict Resolution

When conflicts arise between agents (e.g., competing for the same cell or path):

- Agents SHALL detect potential conflicts based on local perception and received information
- Agents SHALL resolve conflicts using local decision rules such as priority, waiting, or replanning
- Conflict resolution mechanisms SHALL ensure that agents eventually make progress toward their assigned tasks

The system SHALL prevent indefinite blocking situations resulting from unresolved conflicts.

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7. Utility & Success Metrics

7.1 Definitions and Notation

The system operates over discrete time steps $t=1,2,\dots,T$, where T is the total simulation duration.

Let:

- N_{agents} : number of agents
- $N_{completed}$: number of completed tasks
- $t_i^{arrival}$: arrival time of task i
- $t_i^{completion}$: completion time of task i
- D_i^{actual} : distance traveled to complete task i
- $D_i^{optimal}$: shortest – path distance for task i in the static grid
- T_j^{active} : total time agent j is actively executing tasks

7.2 System-Level Metrics

Throughput:

$$Throughput = \frac{N_{completed}}{T}, T > 0$$

Average Task Completion Time:

$$AvgCompletionTime = \frac{1}{N_{completed}} \sum_{i=1}^{N_{completed}} (t_i^{completion} - t_i^{arrival}), N_{completed} > 0$$

Collision Rate:

$$CollisionRate = \frac{N_{collision_events}}{T}, T > 0$$

The target value is zero.

Agent Utilization:

$$Utilization = \frac{1}{N_{agents}} \sum_{j=1}^{N_{agents}} \frac{T_j^{active}}{T}, T > 0$$

Path Efficiency:

$$PathEfficiency = \frac{1}{N_{completed}} \sum_{i=1}^{N_{completed}} \frac{D_i^{optimal}}{D_i^{actual}}, N_{completed} > 0$$

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Deadlock Resolution Time:

$$DeadlockResolutionTime = \frac{1}{N_{deadlocks}} \sum_{k=1}^{N_{deadlocks}} (t_k^{resolved} - t_k^{detected})$$

Congestion Index:

$$CongestionIndex = \frac{1}{T} \sum_{t=1}^T N_{agents}^{congested}(t)$$

where $N_{agents}^{congested}(t)$ is the number of agents located in predefined high-density regions at time t .

7.3 Agent Utility Function

Each agent j is associated with a utility function:

$$U_j = - \left(\alpha \cdot T_j^{task} + \beta \cdot D_j^{travel} + \gamma \cdot C_j^{conflicts} \right)$$

where:

- T_j^{task} : total task execution time
- D_j^{travel} : total distance traveled
- $C_j^{conflicts}$: number of conflicts or delays encountered
- $\alpha, \beta, \gamma > 0$: weighting coefficients

The utility function reflects a cost-minimization objective.

7.4 Social Welfare

The overall system performance is defined as:

$$SW = \sum_{j=1}^{N_{agents}} U_j$$

The system aims to maximize social welfare, which corresponds to minimizing total system cost while maintaining efficient coordination among agents.

7.5 Evaluation Objective

The system is considered successful if it achieves:

- High throughput
- Low average task completion time
- Zero collision rate
- High agent utilization
- Efficient path usage
- Low congestion and fast deadlock resolution

These metrics SHALL be used in Section 8 to evaluate system performance under different scenarios.

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8. Validation Scenarios

The following scenarios are defined to evaluate system performance under controlled and reproducible conditions.

Each scenario specifies setup parameters, evaluation metrics, and expected outcomes.

All scenarios SHALL be executed using a fixed simulation duration T and a configurable random seed to ensure reproducibility.

8.1 Baseline Scenario

Description:

Evaluates system performance under nominal operating conditions.

Setup:

- Number of agents: $N_{agents} = N_{base}$
- Task arrival rate: $\lambda = \lambda_{base}$
- Grid size: fixed
- Simulation duration: T

For example $N_{agents} = 10$, $\lambda = 5 \left[\frac{tasks}{time\ unit} \right]$, $Grid\ Size = 20 \times 20$, $T = 1000$

Evaluation Metrics:

Throughput, AvgCompletionTime, CollisionRate, PathEfficiency

Expected Outcomes:

- CollisionRate = 0
- Stable Throughput
- AvgCompletionTime within acceptable baseline range
- PathEfficiency close to 1

8.2 High Load Scenario

Description:

Evaluates system behavior under high task demand.

Setup:

- $N_{agents} = N_{base}$
- Task arrival rate: $\lambda \gg \lambda_{base}$

Evaluation Metrics:

Throughput, AvgCompletionTime, Utilization, CongestionIndex

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Expected Outcomes:

- Throughput increases up to saturation
- Utilization approaches 1
- AvgCompletionTime increases relative to baseline
- CongestionIndex increases

8.3 High Density Scenario

Description:

Evaluates coordination under high agent density.

Setup:

- $N_{agents} \gg N_{base}$
- Fixed grid size

Evaluation Metrics:

CollisionRate, CongestionIndex, PathEfficiency

Expected Outcomes:

- CollisionRate = 0
- CongestionIndex increases significantly
- PathEfficiency decreases but remains within acceptable bounds
- System maintains continuous progress

8.4 Deadlock Scenario

Description:

Evaluates deadlock detection and resolution capabilities.

Setup:

- Narrow corridors or bottlenecks
- Multiple agents competing for shared paths
-

Evaluation Metrics:

DeadlockResolutionTime, CollisionRate

Expected Outcomes:

- Deadlocks are detected
- DeadlockResolutionTime is finite
- Agents resume task execution after resolution
- CollisionRate = 0

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8.5 Dynamic Environment Scenario

Description:

Evaluates adaptability to changing conditions.

Setup:

- Dynamic task arrival rate (varying over time)
- Localized congestion introduced during runtime

Evaluation Metrics:

Throughput, AvgCompletionTime, CongestionIndex

Expected Outcomes:

- System adapts to changing load conditions
- Throughput remains stable within degradation limits
- AvgCompletionTime increases smoothly (no instability)
- No system halt or deadlock persistence

8.6 Scalability Scenario

Description:

Evaluates system performance as the number of agents scales.

Setup:

- Gradual increase of N_{agents}
- **Fixed grid size and task rate**

Evaluation Metrics:

Throughput, CongestionIndex, Utilization

Expected Outcomes:

- Throughput increases up to a saturation point
- CongestionIndex increases with agent count
- System remains stable (no collisions, no unresolved deadlocks)

8.7 Failure Criteria

A scenario SHALL be considered unsuccessful if any of the following occur:

- CollisionRate > 0
- Deadlock persists without resolution
- System halts or agents cease making progress
- Task completion rate drops to zero under non-zero task availability